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The Expanding Frontier of Agentic AI: A User's View

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The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

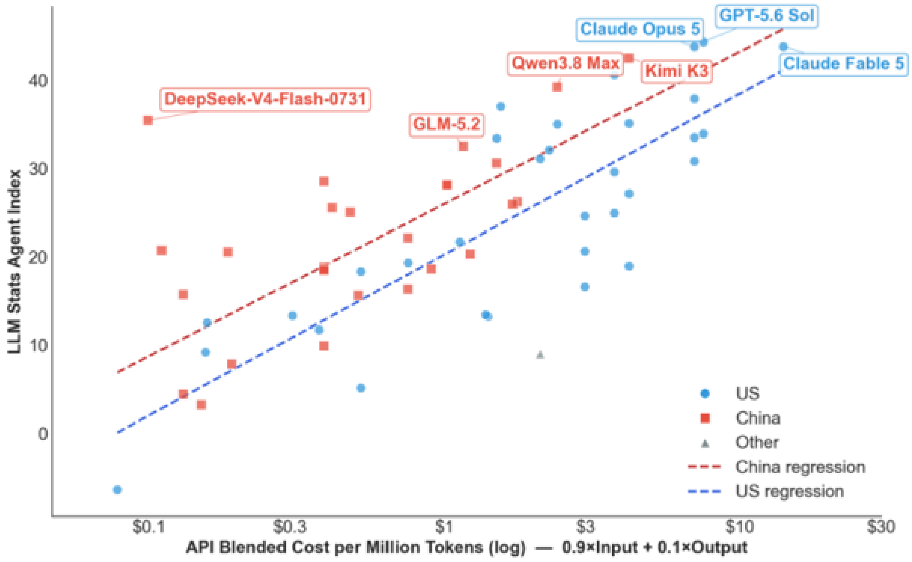
That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the Agent Index from LLM Stats for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

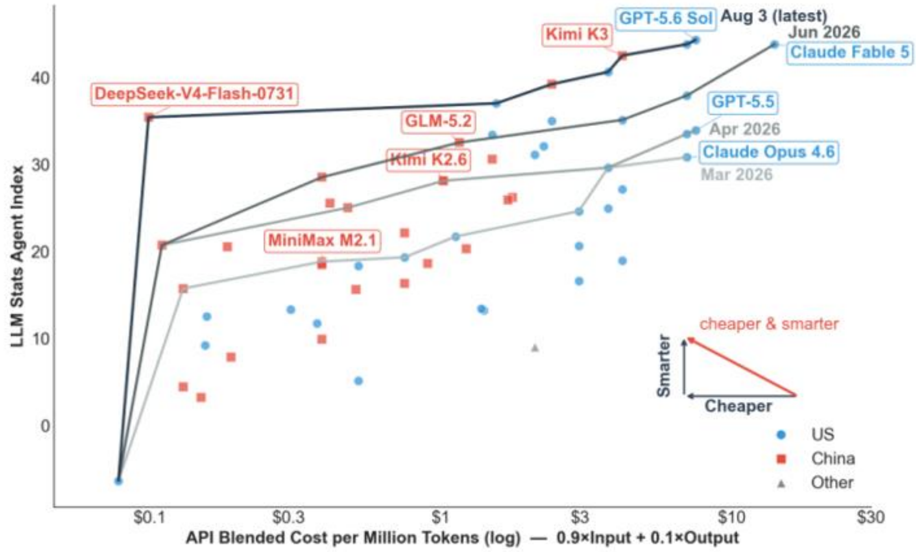
A frontier moving rapidly

The more useful concept for model selection—at least for me as an economist— is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of $7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.

Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of $1.16 is far below Opus 4.7's $7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead… At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

…but Chinese models' advancement from the lower-left is also remarkable. In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at $0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.

DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at $20/month. A few months later, my recurring spending on AI is now reaching $100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an $800 Mac Mini to host my "little lobsters" (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons' paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me smore confident that the future of agentic AI is arriving faster than many expected.**

Appendix 1

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